MIT ACADEMY OF ENGINEERING

DEPARTMENT OF CSE AIML

DSA PRACTICE ASSIGNMENT 1

Name : Nandini Ravindra Chimandare

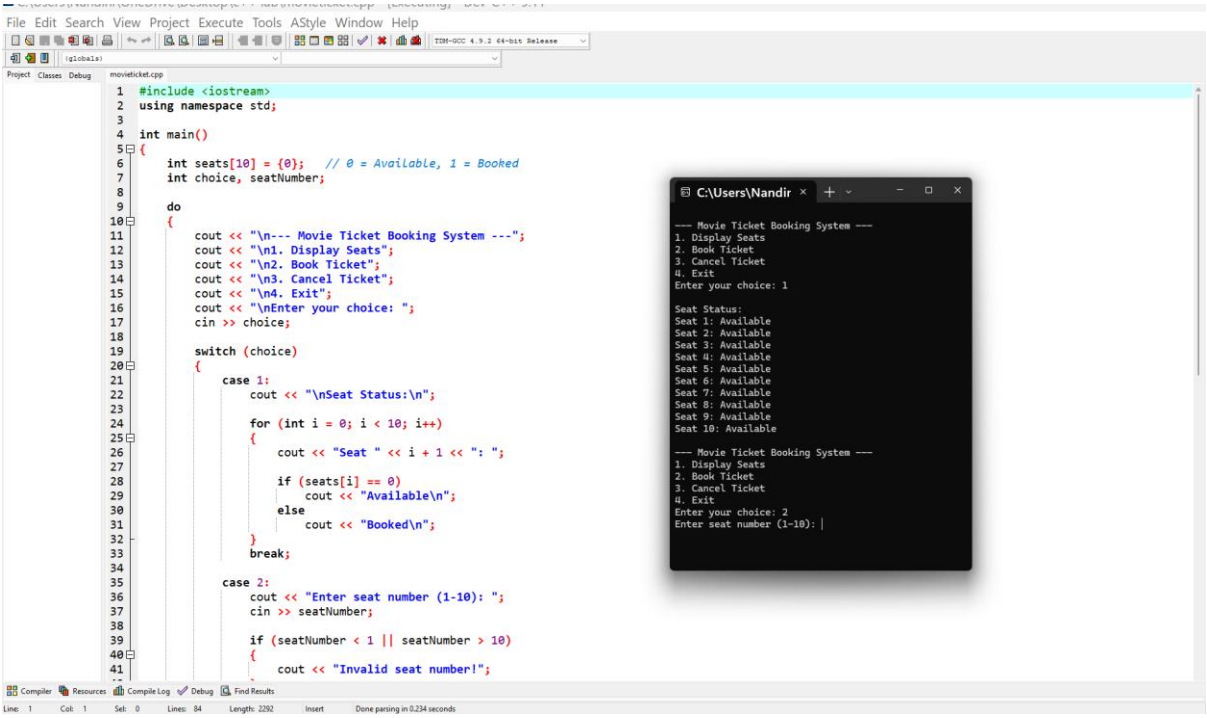
Div/Batch : AIML B1

PRN : 202501110108

Date : 21/08/2026

Topic : Movie Ticket Booking System

Implementation:



Code:

```
#include <iostream>
```

```
using namespace std;

int main()
{
   int seats[10] = {0};   // 0 = Available, 1 = Booked
   int choice, seatNumber;

   do
   {
      cout << "\n--- Movie Ticket Booking System ---";
      cout << "\n1. Display Seats";
      cout << "\n2. Book Ticket";
      cout << "\n3. Cancel Ticket";
      cout << "\n4. Exit";
      cout << "\nEnter your choice: ";
      cin >> choice;

      switch (choice)
      {
         case 1:
            cout << "\nSeat Status:\n";

            for (int i = 0; i < 10; i++)
            {
               cout << "Seat " << i + 1 << ": ";

               if (seats[i] == 0)
                  cout << "Available\n";
               else
                  cout << "Booked\n";
            }
```

```
    break;

case 2:
    cout << "Enter seat number (1-10): ";
    cin >> seatNumber;

    if (seatNumber < 1 || seatNumber > 10)
    {
        cout << "Invalid seat number!";
    }
    else if (seats[seatNumber - 1] == 1)
    {
        cout << "Seat is already booked!";
    }
    else
    {
        seats[seatNumber - 1] = 1;
        cout << "Ticket booked successfully!";
    }
    break;

case 3:
    cout << "Enter seat number to cancel: ";
    cin >> seatNumber;

    if (seatNumber < 1 || seatNumber > 10)
    {
        cout << "Invalid seat number!";
    }
    else if (seats[seatNumber - 1] == 0)
    {
```

```
            cout << "Seat is already available!";
        }
        else
        {
            seats[seatNumber - 1] = 0;
            cout << "Ticket cancelled successfully!";
        }
        break;

    case 4:
        cout << "Thank you!";
        break;

    default:
        cout << "Invalid choice!";
  }

 } while (choice != 4);

 return 0;
}
```